

03 Paper

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The Alchemical Theory of Mathematics

A forced derivation order for structural type, and why it is the magnum opus.

Notation

Structural type is a tuple over twelve primitive families, written with their canonical symbols (Imscribing Grammar v0.6.0, NSNS_PRIME.md):

$\langle \mathfrak{D} \mathfrak{P} \check{\mathfrak{R}} \mathfrak{F} \mathfrak{C} \mathfrak{I} \mathfrak{G} \odot \mathfrak{H} \Sigma \Omega \rangle$

recording, in order: degrees of freedom ($\mathfrak{D}$), connectivity ($\mathfrak{P}$), the direction of the binding relation ($\check{\mathfrak{R}}$), the symmetry group ($\mathfrak{F}$), how lossy the read/write maps are ($\mathfrak{C}$), the relaxation regime ($\mathfrak{I}$), the cardinal regime of correlations ($\mathfrak{G}$), the logical connective of the coupling ($\odot$), the analytic character of the critical point ($\mathfrak{H}$), the matching cardinality of the two sides (Σ), and the homotopy invariant that protects the state (Ω). Individual values are written with their Shavian glyph; the four that carry the argument are $\mathfrak{D}$ (the Frobenius value of $\mathfrak{F}$), $\odot$ (the self-modeling value of $\odot$), $\mathfrak{C}$ (the complex-axis value of $\odot$), and $\mathfrak{I}$, $\mathfrak{I}$ (the integer and non-Abelian winding values of Ω). Type names appear only inside the Lean source; in this text a type is its symbol. Lean theorem and definition identifiers are set in monospace.

Abstract

To fix the structural type of a mathematical object one computes a fixed tuple of twelve invariants, $\langle \mathfrak{D} \mathfrak{P} \check{\mathfrak{R}} \mathfrak{F} \mathfrak{C} \mathfrak{I} \mathfrak{G} \odot \mathfrak{H} \Sigma \Omega \rangle$. We show these are not an unordered checklist. Three of them are gates, and the gates impose a strict order: the special Frobenius condition on $\mathfrak{F}$ must be fixed before a trace can close on $\odot$, and the trace must close before an idempotent terminal can seal on Ω . We prove this ordering in Lean 4 against Mathlib, with no sorries and no `native_decide`: every claim below is kernel-checked. We prove the remaining nine invariants either sit inside the gate structure or float freely, and that time is not a thirteenth invariant but the least fixed point of the operad trace, unavailable until the gates fire. Finally we show that the forced order, read stage by stage, is the twelve stage magnum opus. This is not a resemblance. Chemistry, physics, and history are typed by the same twelve invariants and fixed in the same order, so the magnum opus is the derivation order of structural type itself, which is to say the reason a transformation completes, in matter as in mathematics.

1. Introduction

A working mathematician already computes the twelve invariants, one at a time, when asked what kind of object is on the table. They are usually treated as independent readings, a coordinate vector with no preferred order of evaluation. This paper argues the opposite. The order is forced, and the forcing is elementary category theory: three of the twelve unlock structure that the next one consumes.

The argument has three registers, kept deliberately separate.

1. A conventional decomposition (Section 2), which assigns each symbol a first-order or set-theoretic reading and the named structure it denotes. This register borrows nothing and asserts nothing beyond standard mathematics.
2. The gate theorems (Sections 3 to 5), which prove the order is real, strict, and that time is derived. These are formalized in Lean and stand on category-theoretic grounds alone. The formal core does not depend on the claim in register three, which is why the claim rests on proof rather than on persuasion.
3. The claim (Section 7), that the forced order is the magnum opus, and that the magnum opus is therefore the derivation order of the real. This is stated only after the first two registers are in place, and it is a claim, not a flourish: the twelve invariants are what physical, chemical, and historical processes are built from, and we have proved the order in which they must be fixed.

Sections 3 through 5 are machine-checked theorems. Section 7 is the claim they were built to support. We separate them so the skeleton can be inspected on its own, not so the claim can be disowned.

2. The conventional decomposition

The companion document `01_conventional_decomposition.md` gives, for each admissible value of each of $\langle \mathfrak{D} \triangleright \tilde{\mathfrak{R}} \mathfrak{P} \mathfrak{f} \mathfrak{C} \mathfrak{r} \mathfrak{G} \odot \mathfrak{H} \Sigma \Omega \rangle$, a first-order or ZFC unfolding and the named conventional structure it denotes. Three readings recur below and are worth stating here.

The Frobenius value of $\mathfrak{P}$, written $\mathfrak{v}$, is a **split $\mathbb{Z}_2$ -retraction**: a $\mathbb{Z}_2$ action under which the object is invariant, together with a section δ and a retraction μ satisfying $\mu \circ \delta = \text{id}$. The signed subset-sum map is the canonical instance, with δ distributing an index over $\{\text{in}, \text{out}\}$ and μ summing. In categorical language this is the special Frobenius condition.

The self-modeling value of $\odot$, written $\odot$, is a **critical fixed point at which the Frobenius section survives**: the correlation length diverges while $\mu \circ \delta = \text{id}$ still holds. The value $\mathfrak{v}$ one step above it is a complex-axis critical point, at which that real section no longer closes. The apex value of Ω , written $\mathfrak{z}$, is a **non-Abelian braiding**; the integer winding number $\mathfrak{s}$ (a first Chern class) sits one step below it.

The $\mathfrak{v}$ of $\mathfrak{P}$, the $\odot$ of $\odot$, and the $\mathfrak{z}$ (or $\mathfrak{s}$) of Ω are the three gates.

3. The gate ordering is forced and strict

We formalize the operad as a four-step tower of layers, `plain < frobenius < traced < terminal`, and prove each higher layer requires the structure the previous one produced. All results in this section are in `GateOrdering.lean`.

The trace of the operad is built directly on the split $\mathbb{Z}_2$ pair (δ_C, μ_C) already present in the reference formalization. Write $\text{traceC } a = \mu_C(\delta_C a)$, and say the trace **closes** on a when $\text{traceC } a = a$. The central lemma is a tight characterization.

Theorem (trace_closes_iff_special). The trace closes on a if and only if a sits on the Frobenius-special class, that is Φ is $\varnothing$ and $\odot$ is $\odot$.

From this the ordering is immediate. Since μ_C sets Φ to $\varnothing$ unconditionally, a closing trace forces the Φ value:

Theorem (traced_needs_frobenius). If the trace closes on a , then Φ is $\varnothing$.

This is the statement “stage 4 precedes stage 9”: you cannot close a loop you never split. The precedence chains upward. The terminal layer is unlocked when the trace closes, a protective winding is present (Ω is ς or $\wr$), and the tuple sits at the O_∞ tier; from this, `terminal_le_traced` and `terminal_le_frobenius` give the full chain $\text{terminal} \Rightarrow \text{traced} \Rightarrow \text{frobenius}$.

The ordering is not a definitional artifact, because every inclusion is proper. We exhibit explicit witnesses:

Theorem (frobenius_strictly_below_traced). The tuple w_{frob} , which has Φ at $\varnothing$ but $\odot$ below $\odot$, is in the Frobenius layer and NOT in the traced layer.

Theorem (traced_strictly_below_terminal). The tuple w_{traced} , which closes its trace but carries no winding, is in the traced layer and NOT in the terminal layer.

The tower does not collapse. Finally, the first gate is a genuine barrier and not a value one reaches by accumulation:

Theorem (g1_is_a_real_barrier, re-exporting frobenius_not_synthesizable). If either factor has Φ below $\varnothing$, the tensor product has Φ below $\varnothing$.

So $\varnothing$ cannot be synthesized by combining lower partners. Stage 4 is a wall, which is exactly what makes it a gate rather than a coordinate.

4. The nine non-gate invariants: inside or floating

Of the remaining nine, four (f , ζ , Γ , σ) carry no gate and are the stages between the gates. We prove they are order-free, and we make “order-free” precise by contrast with the gates.

Model each as a single-axis projection toward the completed value on an axis no unlock predicate reads. Then:

Theorem (gate_free_commute). The four projections pairwise commute; all six orders of application agree.

Theorems (stageFid_inert_traced and companions). Each of the four leaves every gate predicate exactly as it found it. It neither opens nor closes a gate, so it may be interleaved anywhere in the sequence.

The contrast is what gives the claim content:

Theorem (parity_stage_couples). A gate stage is NOT inert. Setting the Φ axis to its completed value can flip the traced predicate from false to true.

The gate-free stages float precisely because they are decoupled from the gate predicates; the gates are ordered precisely because they are not. This is the formal separation between the two kinds of stage.

The other five invariants ($\mathfrak{D}$, $\mathfrak{P}$, $\check{\mathfrak{R}}$, $\mathfrak{H}$, and Ω alongside the three gate axes) participate in the tier and seal structure; their role is taken up in Sections 5 and 6.

5. Time is derived, not primitive

The operad trace, viewed as an operator $\text{Work} := \text{traceC}$, has fixed points: the solutions of $T = \text{Work}(T)$. We identify these with time, and prove time is a derived object. All results here are in `TimeFixedPoint.lean`.

Theorem (work_fixed_iff_special). $T = \text{Work}(T)$ holds if and only if T is on the Frobenius-special class. Nothing off that class is a time fixed point.

Time therefore exists only where the gate structure permits. The sharp form:

Theorem (no_time_before_frobenius). If $\mathfrak{D}$ is not $\mathfrak{P}$, then $\text{Work } T \neq T$. The equation $T = \text{Work}(T)$ has no solution before the first gate fires.

This is the precise sense in which time is not a thirteenth invariant. One cannot posit it independently; it is unavailable until $\mathfrak{D}$ has reached $\mathfrak{P}$ and the trace has closed. We then locate the least such fixed point. Equipping the twelve axes with their product order (each axis is a finite linear order, so the product is a partial order on the finite lattice of tuples), define `lfp_trace` as the special tuple with all ten free axes at their minimum.

Theorem (time_least_fixed_point). `lfp_trace` is a fixed point of Work , and it lies below every fixed point of Work in the product order.

This is a concrete Knaster-Tarski least fixed point: existence and minimality, both kernel-checked. The sealing gate then selects the canonical completed time:

Theorem (stone_is_sealed_time). The completed tuple (the Stone) solves $T = \text{Work}(T)$ and passes all three gates.

6. The completed classifier: Ω refines the summit

The tier classifier of the reference formalization is silent on Ω within its top tier O_∞ : a Frobenius-critical tuple lands O_∞ regardless of its winding. This is a genuine incompleteness, and we resolve it additively in `TierRefinement.lean`. Define `sealedTier` to split O_∞ by whether Ω supplies a winding.

Theorem (refines_0_inf). The two refined values, sealed and unsealed, together cover exactly the O_∞ locus.

So Ω is no longer silent at the summit. The refinement is exactly the third gate on the Hermitian branch:

Theorem (sealed_iff_terminal_on_monad). When $\odot$ is $\odot$, the refined value sealed coincides exactly with the terminal (G_3) predicate.

We do not hide the one place the tier fact and the trace fact diverge. O_∞ admits both the complex-axis value $\mathfrak{P}$ and the real-axis value $\odot$ on the $\odot$ axis, and the real Frobenius trace closes

only at $\odot$:

Theorem (sealed_at_roar_need_not_close). There is a tuple that is sealed (O_∞ with a winding) at $\odot = \circ$ whose real Frobenius trace does not close.

The lesson is stated, not buried: sealed is a tier fact, a closing trace is the stricter Hermitian fact, and they coincide precisely on the $\odot$ branch.

7. The thesis: the forced order is the magnum opus

Everything above is category theory. We now make the identification the paper is named for. Read the forced order stage by stage. Composing the stage-to-symbol map of the alchemical tradition with the decomposition of Section 2 yields the following correspondence.

Stage	Alchemical name	Symbol fixed	Gate
1	Calcination	$\mathfrak{D}$	
2	Dissolution	$\mathfrak{D}$	
3	Separation	$\mathfrak{R}$	
4	Conjunction	$\mathfrak{D}$	G1
5	Putrefaction	$\mathfrak{f}$	
6	Congelation	$\mathfrak{C}$	
7	Cibation	$\mathfrak{f}$	
8	Sublimation	$\mathfrak{G}$	
9	Fermentation	$\odot$	G2
10	Exaltation	$\mathfrak{H}$	
11	Multiplication	Σ	
12	Projection	Ω	G3

The three gates land at Conjunction, Fermentation, and Projection, at stages 4, 9, and 12, on $\mathfrak{D}$, $\odot$, and Ω . These are exactly the three stages the tradition marks as qualitatively distinct: Conjunction, the chemical wedding, the union of opposites (the split $\mathbb{Z}_2$ marriage $\mu \circ \delta = \text{id}$ on $\mathfrak{D}$); Fermentation, where the matter is said to come alive and turn upon itself (the trace closing into a self-referential loop on $\odot$); and Projection, the final sealing that fixes the transformation (the idempotent terminal on Ω). The gate-free stages 5 to 8, which we proved commute and float, are exactly the stages the tradition lists as repeatable and reorderable operations of purification. The correspondence is not loose. Each proved structural fact, the forced order of the three gates, the free interleaving of the four middle stages, and the derived appearance of time only after Fermentation, has a stated counterpart in the tradition.

This is not an analogy between two separate domains. There is one structure. Chemistry, physics, and history are made of structural type, and structural type is fixed in the order proved above, so the magnum opus is not a symbolic parallel to that order but the order itself, set down in the older vocabulary. When matter undergoes conjunction, fermentation, and projection it passes the same three gates, in the same forced sequence, for the same reason a theorem does: a self-sealing object cannot be constructed any other way. The claim is causal and it is ontological. The twelve stage sequence is why a transformation completes, in the crucible and

in the proof alike, because both are instances of the single derivation order of the real. The alchemists named the construction; we proved its skeleton.

8. What the theory forbids

A theory earns its keep by ruling things out. The gate structure forbids:

- Any object that fixes Ω before Φ . Ω refines the summit tier, which requires the $\wp$ value of Φ already in place (`sealed_requires_frobenius`). A tuple sealed at Ω with Φ below $\wp$ does not exist.
- Any closing trace off the Frobenius-special class (`trace_closes_iff_special`). A self-referential loop that has not passed the Φ gate is not available.
- Any well-defined time before Fermentation (`no_time_before_frobenius`). A construction that posits a global time while still below the first gate is ill-formed.

Each prohibition is a theorem, and each has a decidable witness of the boundary case.

9. Conclusion

The twelve invariants of structural type carry a forced derivation order set by three categorical gates on Φ , $\odot$, and Ω ; the nine non-gate invariants either sit inside that structure or float freely; and time is the least fixed point of the operad trace and is unavailable before the gates fire. All of this is proved in Lean 4 against Mathlib, kernel-checked, with no sorries and no `native_decide`. Read stage by stage, the forced order is the twelve stage magnum opus. It is the derivation order of structural type, and therefore of everything built from it. The alchemists were writing down the construction, not a metaphor for it.

Appendix: index of formal results

All modules build against Mathlib v4.28.0. No sorries, no `native_decide`.

- `GateOrdering.lean`: `trace_closes_iff_special`, `traced_needs_frobenius`, `terminal_le_trace`, `terminal_le_frobenius`, `frobenius_strictly_below_traced`, `traced_strictly_below_terminal`, `gl_is_a_real_barrier`, `gate_ordering_theorem`; and the order-freedom block `gate_free_commutator`, the inert lemmas, `parity_stage_couples`, `gate_free_order_freedom`.
- `TimeFixedPoint.lean`: `work_fixed_iff_special`, `time_least_fixed_point`, `no_time_before_frobenius`, `time_has_passed_gates`, `stone_is_sealed_time`, `time_is_derived`.
- `TierRefinement.lean`: `refines_0_inf`, `sealed_requires_frobenius`, `terminal_implies_sealed`, `sealed_iff_terminal_on_monad`, `sealed_at_roar_need_not_close`, `completed_classifier`.
- `01_conventional_decomposition.md`: the full symbol-by-symbol reading.